Report of CSC8645 - Assignment on Food Segmentation and Multimodal Idiom Understanding

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I. Task 1: Food Image Segmentation

A. Introduction

Food image segmentation focuses on identifying and labeling each pixel in an image according to different food categories. This task is more challenging than image classification because multiple food items can appear together with similar colors and textures. Models such as U-Net [1] and DeepLabV3+ [2] have been widely used for segmentation, while more recent transformer-based models such as SegFormer [3] aim to improve performance by capturing global context.

In this task, we implemented and compared different segmentation models in the FoodSeg103 dataset to understand how model design affects performance.

B. Methodology

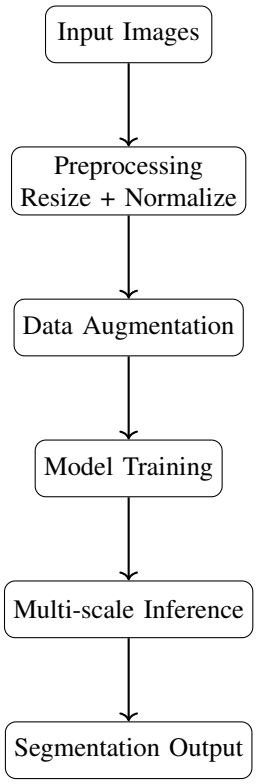

Fig. 1. Overall segmentation pipeline used in this task.

1) Pipeline Overview: The dataset was loaded into colab environment via a zip file and unziped onto to the colab environment then all required libraries were downloaded or imported into the notebook. The images were resized and normalized using ImageNet statistics to ensure compatibility with pretrained encoders. Data augmentation techniques such as random flipping, cropping and colour variations were applied to improve generalization and reduce overfitting.

Three models were implemented progressively to analyse performance improvements. First, a U-Net model with a ResNet34 encoder was used as a baseline, providing a simple encoder–decoder structure for pixel-wise prediction. Next, DeepLabV3+ with ResNet50 and ResNet101 encoders was implemented to capture multi-scale contextual information using atrous convolution and spatial pyramid pooling. Finally, SegFormer-B3 was introduced as a transformer-based model capable of modelling long-range dependencies across the image.

During evaluation, multi-scale inference was applied by generating predictions at different image resolutions (0.75×, 1.0×, 1.25×) and averaging the outputs. This approach helps reduce scale sensitivity and improves overall segmentation accuracy.

C. Results

TABLE I
Segmentation Performance on Test Set

Model	mIoU (%)
U-Net (ResNet34)	14.74
DeepLabV3+ (ResNet50)	17.63
DeepLabV3+ (ResNet101)	21.27
SegFormer-B3	31.50
SegFormer + Multi-scale	**33.57**

D. Discussion

The results show a steady improvement in performance as more advanced models are used. The U-Net model provides a basic level of segmentation but struggles with detailed boundaries. DeepLabV3+ improves performance by capturing multi-scale features, but still misses finer details in some cases.

The SegFormer model(B3) performs significantly better, producing more accurate and consistent segmentation masks.

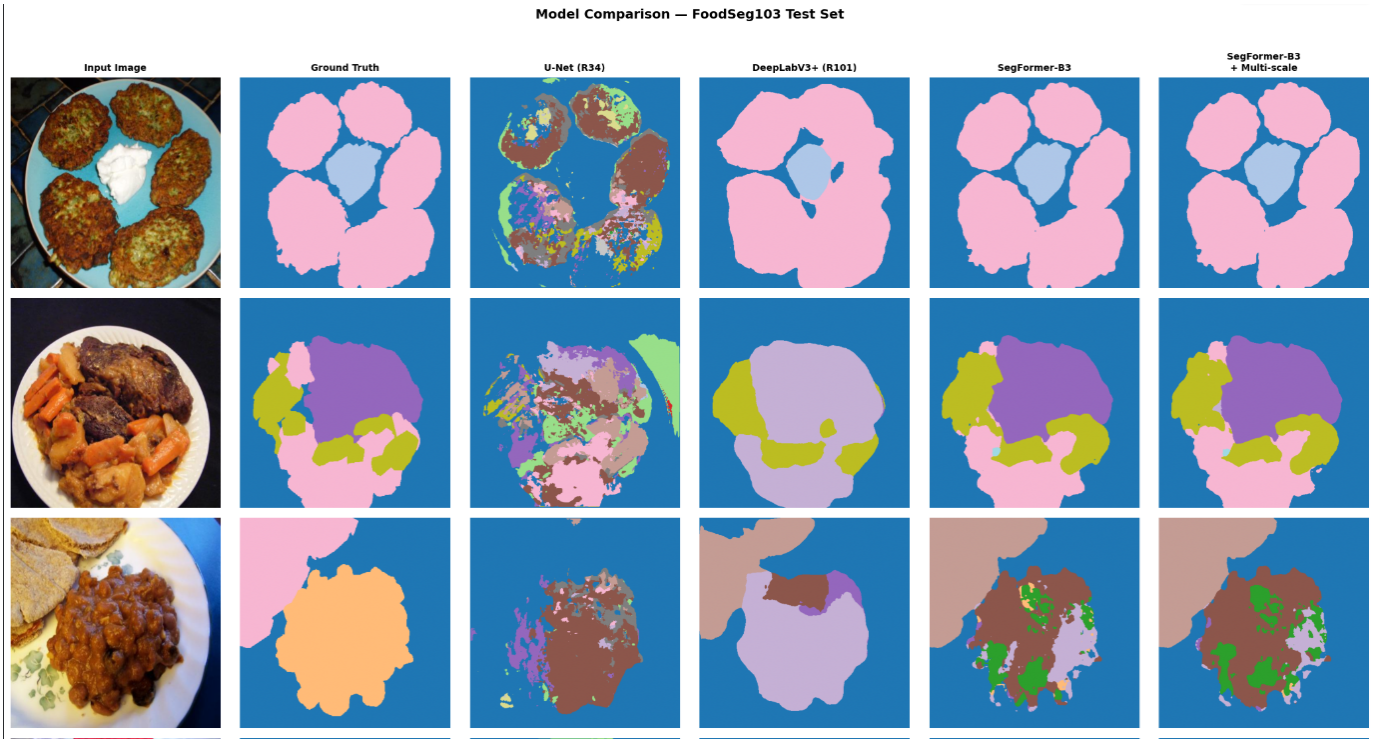

Fig. 2. Qualitative comparison of segmentation models on the FoodSeg103 dataset.

This suggests that modelling global context is important for distinguishing similar food items.

Another important observation was the impact of preprocessing. Using ImageNet normalisation improved performance noticeably, highlighting the importance of matching pretrained model expectations.

Multi-scale inference also provided a small but consistent improvement without requiring additional training.

E. Conclusion

This task shows that more advanced architectures improve segmentation performance. U-Net provides a baseline, while DeepLabV3+ enhances results through multi-scale feature extraction. SegFormer achieves the best performance, highlighting the importance of global context modelling in complex images. Additionally, preprocessing steps such as normalisation and techniques like multi-scale inference further contribute to improved accuracy and more consistent segmentation results.

II. Task 2: Multimodal Idiom Understanding

A. Introduction

This task focuses on understanding idiomatic expressions using both text and images. Idioms are challenging because their meanings are often figurative rather than literal. We used the CLIP model for this task which [4] can match text and images, but it is mainly trained on literal descriptions.

In this task, we evaluate how well CLIP performs on idiomatic expressions and how fine-tuning improves its performance.

B. Methodology

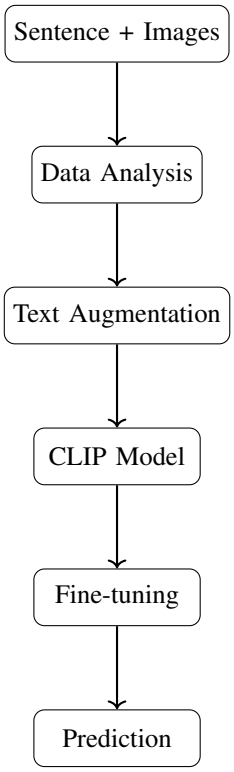

Fig. 3. Pipeline for multimodal idiom understanding.

1) Pipeline Overview: The dataset consists of sentences paired with three candidate images, where only one image correctly represents the meaning of the sentence. Exploratory data analysis confirmed the absence of missing values and duplicates, while also revealing variability in sentence length and structure, reflecting different contextual uses of idiomatic expressions.

To enhance the dataset, text augmentation was performed using large language models to generate paraphrased versions of the original sentences while preserving their semantic meaning. This increased linguistic diversity and helped the model generalise better. A zero-shot CLIP model was initially used, where cosine similarity between text and image embeddings was computed to select the most relevant image.

The model was then fine-tuned using labelled data in a supervised setting. Each training instance consisted of one sentence and three candidate images, and the model was trained to assign the highest similarity score to the correct image. This allowed the model to learn task-specific relationships between textual context and visual representations.

C. Results

TABLE II
CLIP PERFORMANCE

Model	Accuracy
CLIP (Zero-shot)	11%
CLIP (Fine-tuned)	**89%**

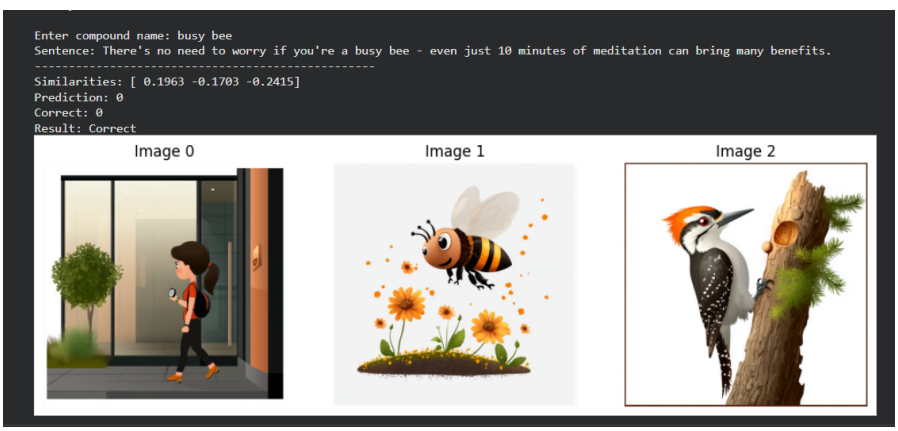

Fig. 4. Example prediction for an idiomatic expression after fine-tuning.

D. Discussion

The zero-shot CLIP model performed poorly because it relies on literal relationships between text and images. Idiomatic expressions require understanding context and meaning beyond direct visual cues.

After fine-tuning, the model achieved much higher accuracy. This shows that training on task-specific data helps the model learn how idioms relate to visual representations.

Text augmentation also helped improve performance by exposing the model to different ways of expressing the same meaning.

E. Conclusion

This task demonstrates that pretrained models such as CLIP are not sufficient for handling complex, non-literal language tasks in a zero-shot setting. Fine-tuning allows the model to learn task-specific relationships between text and images, significantly improving performance. Additionally, data augmentation enhances robustness by introducing linguistic diversity, leading to better generalisation and more accurate multimodal predictions.

III. REFERENCES

REFERENCES

[1] O. Ronneberger et al., U-Net, 2015.
[2] L.-C. Chen et al., DeepLabV3+, 2018.
[3] E. Xie et al., SegFormer, 2021.
[4] A. Radford et al., CLIP, 2021.
[5] FoodSeg103 Dataset, Dataset Link, 2021.